

# Alexander Belik

Software Engineering | AI Systems | Scientific Computing

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## PROFILE

Theoretical Physics student and software builder focused on Python automation, AI systems, scientific ML, and practical full-stack tools.

I build software for messy real-world workflows: typed boundaries, reproducible data pipelines, inspectable automation, and interfaces people can actually use.

## CORE STACK

Python, TypeScript, React  
FastAPI, PyQt, SQLite  
NumPy, SciPy, PyTorch  
QuTiP, Optuna, Matplotlib  
Selenium, Docker, Git  
LaTeX, Markdown, Figma

## ENGINEERING AREAS

Backend APIs and services  
Data scraping and normalization  
Workflow automation  
Local LLM orchestration  
Tool routing and guarded actions  
Scientific simulation  
ML diagnostics and tuning

## EDUCATION

### B.A. Theoretical Physics

Trinity College Dublin, expected 2026

Focus: computational physics, quantum mechanics, statistical mechanics, and QFT.

### Leaving Certificate

The Dublin Academy of Education,  
2020–2022

## RESEARCH BASE

Numerical simulation  
Data analysis  
Reservoir computing  
Quantum dynamics  
Experiment-control tooling  
Technical writing

## LANGUAGES

English – fluent  
Russian – intermediate  
Spanish – basic

## EXPERIENCE

### Software Engineer, EdTech

2024–2025

Grinds360

- Built Python automation tools for video, content, and interactive lesson workflows.
- Contributed to full-stack product features and UX iteration with Figma-backed collaboration.
- Translated ambiguous product requirements into maintainable tools with non-technical stakeholders.

### Research Assistant, CRANN Institute

Jun–Sep 2025

Trinity College Dublin

- Built a PyTorch Echo State Network toolkit for quantum time-series prediction.
- Added configurable reservoirs, mixed precision, ridge-regression solvers, Optuna tuning, logging, and diagnostics.
- Validated predictions against Heisenberg spin-chain simulation data and conservation-law checks.

### Experimental / Computational Research Assistant

May–Sep 2024

Trinity College Dublin

- Created Python control and data-acquisition tooling for spectroscopy experiments on ferrofluids and liquid crystals.
- Improved repeatability through structured logging and automated measurement workflows.

### Teaching Assistant

Oct 2025–Present

Trinity College Dublin

- Lead weekly tutorials, support problem solving, and grade assignments with clear feedback.

## SELECTED PROJECTS

### ORION / Big Brother Assistant

Python, PyQt, Ollama, SQLite, Whisper

Windows-first local assistant with wake-word gating, LLM-first capability selection, memory, guarded tools, daemon/direct/offline modes, and PyQt overlay/dashboard surfaces.

### Choose Car

React, TypeScript, FastAPI, SQLite, Selenium

Ireland-focused keep-vs-switch car optimizer with backend services, admin review tools, OEM and used-inventory scraping, robots-aware indexed discovery, and normalization pipelines.

### EchoState and Heisenberg Chain

PyTorch, NumPy, SciPy, QuTiP, Optuna

Reusable ESN library plus quantum spin-chain simulator for learning physical dynamics from observables, with solver comparisons and stability diagnostics.

### DFT Hamilton / PySCF Environment

Docker, Python, PySCF, LibXC

Reproducible computational chemistry workflow for H2 RHF/DFT, open-shell Li UHF, molecular examples, spin-correlation tests, endpoint sensitivity checks, and custom functional comparisons.

## SOFTWARE STRENGTHS

**Practical architecture:** separates orchestration, persistence, UI, and side effects into testable boundaries.

**Data-heavy workflows:** comfortable with scraping, raw/normalized databases, review surfaces, and repeatable imports.

**Scientific reliability:** uses diagnostics, conservation checks, reproducible environments, and structured logs to make model behavior inspectable.